When Proximal Pessimism Cannot Contract: A Schedule-Free Floor and Two Checkable Switches

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Abstract

Offline RL in confounded POMDPs identifies policy values through *bridge functions* and then plans pessimistically over a confidence region around the fitted bridge. The construction is only useful if that region contracts as data accumulates. We show it need not. For any ridge $\lambda > 0$ and *any* width rule — power-law, logarithmic, data-adaptive, even truth-dependent — a region that covers the truth pays a penalty of at least $\beta\beta_g$ at every sample size, where β is the true bridge's mass in the design's null space and β_g the value gradient's. The regularizer cancels, so no schedule escapes it. We then characterize exactly when this floor is nonzero, and the answer is a conjunction of two conditions a practitioner can check *before* fitting: an inadequate negative control ($|O_0| < |S|$, a statement about dimensions alone) makes the truth leak, and confounding — the behaviour policy depending on latent state — makes the gradient leak. Neither alone suffices; we verify the first in $50/50$ population cells and map the second across four configurations. The characterization concerns the *first* stage: from $t = 2$ on, history restores per-action completeness, so exactly one block leaks whatever the horizon and the floor does not compound. Placing the schedule of Hong et al. (2024) inside this family shows its governing exponent is positive for *every* admissible smoothness ($30/30$ cells), so the region it prescribes is non-contracting on the leaky branch. At the decision level, over 20 seeds, pessimistic and plug-in selection have opposite-signed regret slopes under that schedule and their curves cross: pessimism helps at $N{=}2{,}000$ and hurts at $N{\geq}32{,}000$. That consequence, however, is short-horizon: the decision-relevant value spread grows with T while the floor does not, so the fraction of the selection problem it distorts decays as $1/T$. The width result is horizon-free; its decision-level consequence is not. All of this lives outside the anchor paper's own assumptions, which is what makes it a robustness result about the boundary of applicability rather than a defect.

1. Introduction

Offline reinforcement learning from logged data is biased when the logging policy acted on latent state the learner never sees. Proximal causal inference gives a principled route out: with a *negative control* variable, reward and transition models are identified through bridge functions solving conditional moment restrictions (Miao et al., 2018; Tchetgen Tchetgen et al., 2020), and this has been carried into sequential decision making both model-free (Shi et al., 2022; Bennett & Kallus, 2024) and model-based (Hong et al., 2024).

Because the identified object is estimated, these methods plan *pessimistically*: they build a confidence region around the fitted bridge $\hat{b}$ and evaluate a policy at its worst admissible member. For a value functional linear in the bridge block, the region is an ellipsoid and the inner minimum has the closed form

$$V_{\text{low}} = \langle g, \hat{b} \rangle - W, \qquad W = \sqrt{\xi}\, \|g\|_{H^{-1}}, \qquad (1)$$

with $H = \hat{T}_2 + \lambda I$ the ridged stage-2 design and ξ the region radius. The whole construction rests on a premise that is rarely stated: that W shrinks as N grows, so the pessimistic value approaches the true one and selection eventually becomes correct.

This paper. We show the premise can fail, characterize exactly when, and locate an existing method inside that characterization.

1. **A schedule-free floor** (Section 3). Any width rule whose region covers the truth satisfies $W \geq \beta\beta_g$ at every N, where β and β_g are the null-space masses of the true bridge and of the value gradient. The proof is two lines and the regularizer cancels, so this is not a

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statement about a particular schedule — it holds for all of them.

2. **Two checkable switches** (Section 4). The floor is nonzero exactly when *both* an inadequate negative control and genuine confounding are present. Incompleteness makes the truth leak but leaves the gradient inside the identified span; confounding makes the gradient leak but, under completeness, leaves the truth inside. Both conditions are assessable in advance, one from dimensions alone.
3. **Where an existing schedule sits** (Section 5). Writing the schedule of Hong et al. (2024) in our parametrization, its governing exponent is positive for every admissible smoothness parameter, so on the leaky branch the penalty it assigns *grows* with N.
4. **A decision-level consequence** (Section 6). In a design where both switches are on, over 20 seeds, pessimistic and plug-in regret slopes have opposite signs and the two curves cross.

What we are not claiming. Section 7 is explicit that the binding hypothesis fails under the anchor paper's own assumptions: completeness forces $\beta = 0$, and zero population gradient leakage is equivalent to a finite concentrability coefficient by that paper's own Lemma C.1. Everything here is therefore a statement about what happens *outside* those assumptions. That is the right question for a robustness study, and we label it as one rather than presenting it as a flaw.

2. Setting

We work per bridge block. The stage-2 empirical risk is exactly quadratic, so the confidence region is an exact ellipsoid,

$$\mathrm{conf}(\xi) = \{\, b : (b-\hat{b})^\top H(b-\hat{b}) \le \xi \,\}, \quad H = \hat{T}_2 + \lambda I, \tag{2}$$

and Equation (1) is the standard support-function minimum. Write Nul for the null space of $\hat{T}_2$, P_{Nul} for the orthogonal projector onto it, and

$$\beta := \|P_{\mathrm{Nul}} b_\star\|, \qquad \beta_g := \|P_{\mathrm{Nul}} g\|. \tag{3}$$

Assumption 1 (Exact structural null)**.** $\hat{T}_2$ has an exact null space with $\dim \mathrm{Nul} \ge |O| - \min(|O|, |O_0|) > 0$, fixed by the dimensions of the observation and negative-control alphabets and independent of N.

Assumption 1 is a property of a bridge parametrized over the observation space but identified only through the latent bottleneck: when the observation alphabet is richer than the instrument can resolve, the design matrix is structurally rank-deficient at every sample size. It is exact in the tabular case because the design is a finite-rank count matrix; Section 7 discusses what survives for a strictly positive-definite kernel.

We further assume the fitted bridge is the minimum-norm solution, so $P_{\mathrm{Nul}}\hat{b} = 0$. The ridge solve satisfies this exactly, to machine precision, on both solver paths we implement.

3. A Schedule-Free Floor

Proposition 1 (Floor)**.** *Under Assumption 1 and $P_{\mathrm{Nul}}\hat{b} = 0$, for* any *ridge $\lambda > 0$ and* any *width $\xi > 0$ — power-law, logarithmic, data-adaptive, or chosen with knowledge of the truth — a region satisfying $b_\star \in \mathrm{conf}(\xi)$ obeys*

$$W \;\ge\; \beta\,\beta_g \qquad \text{at every } N. \tag{4}$$

Proof. Coverage of the truth constrains the radius from below. Decomposing $b_\star - \hat{b}$ and keeping only its null component, on which H acts as λI,

$$\xi \;\ge\; (b_\star - \hat{b})^\top H (b_\star - \hat{b}) \;\ge\; \lambda\beta^2.$$

Separately $H^{-1} \succeq \lambda^{-1} P_{\mathrm{Nul}}$, so $\|g\|_{H^{-1}} \ge \beta_g/\sqrt{\lambda}$. Multiplying,

$$W = \sqrt{\xi}\,\|g\|_{H^{-1}} \;\ge\; \sqrt{\lambda}\,\beta \cdot \frac{\beta_g}{\sqrt{\lambda}} \;=\; \beta\beta_g. \qquad \square$$

The regularizer enters both factors and cancels. That is why the statement is schedule-free: tuning λ trades the coverage requirement against the gradient norm at exactly the rate that leaves their product fixed. A tighter region buys a smaller $\|g\|_{H^{-1}}$ only by demanding a larger ξ to keep covering the truth.

What can escape it. Proposition 1 is not a claim that pessimism is hopeless. Three routes out are visible in the proof, and none of them is a width rule: centring the region somewhere other than the min-norm solution (which breaks $P_{\mathrm{Nul}}\hat{b} = 0$), using a non-ellipsoidal region (which breaks the support identity), or arranging $\beta\beta_g = 0$. The third is the practical one, and Section 4 is about when it holds.

4. Two Checkable Switches

The floor is a product. It vanishes if either factor does, so the useful question is not "is the design ill-posed" but "which of the two leakages is present". They turn out to be controlled by different, independent properties of the problem. All results in this section are exact population algebra — no sampling, no rank selection, no estimation noise.

4.1. Incompleteness leaks the truth

The identification argument requires a completeness condition: the negative control must resolve the latent state.

Table 1. Population null-space share of the true bridge, β, as a fraction of $\|b_\star\|$, across configurations and confounding levels. Exact algebra, no sampling.

$(\lvert S\rvert, \lvert O\rvert, \lvert O_0\rvert)$	0.0	0.3	0.6	0.9	1.0
$(2, 6, 4)$	0	0	0	0	66.0%
$(2, 3, 2)$	0	0	0	0	65.7%
$(3, 7, 5)$	0	0	0	0	56.9%
$(2, 6, 2)$	0	0	0	0	64.3%
$\mathbf{(4, 6, 2)}$	**71.3%**	**71.3%**	**71.1%**	**69.4%**	**69.5%**

When it does not — concretely when $|O_0| < |S|$ — the minimum-norm bridge acquires mass in the design null.

We evaluate β in the population over five configurations and ten confounding levels. The prediction holds in $50/50$ **cells**, and the dichotomy is *sharp* rather than graded: $\beta \leq 1.5 \times 10^{-15}$ in every complete cell and between 56.9% and 77.3% of the bridge norm in every incomplete one, with nothing in between (Table 1; maximum bridge residual 4.4×10^{-16}, so the min-norm solve is exact throughout).

Two routes therefore produce $\beta > 0$, and they are not equally useful. Saturating the confounding knob at exactly 1.0 works in a complete design, but it is a knife-edge: at 0.9 completeness is restored and β returns to machine zero, and a deterministic behaviour policy is contrived. The dimensional route $|O_0| < |S|$ is the robust one — it holds at *every* confounding level, as the last row of Table 1 shows, and it is checkable before fitting anything.

4.2. Confounding leaks the gradient

We initially assumed β_g travels with β. It does not, and the distinction is the substance of this section.

The stage-1 gradient's observation profile is the marginal $\nu_1 = E^\top p_1$, where E is the emission matrix and p_1 the initial state distribution. The design span for action a is generated by $E^\top v_{a,x}$ with $v_{a,x}[s] \propto p_1[s]\, \pi_b[s,a]\, K_0[s,x]$. If π_b does *not* depend on s, then $\pi_b[s,a]$ factors out and summing over x returns $E^\top p_1$ up to scale. Hence:

Proposition 2 (Unconfounded gradients cannot leak)**.** *If the behaviour policy is independent of the latent state, then $\nu_1 \in \mathrm{span}\{E^\top v_{a,x}\}_x$ and hence $\beta_g = 0$, however incomplete the negative control is.*

Proposition 2 carries one dependency worth stating: the argument needs K_0 row-normalized. Were the negative control subject to state-dependent missingness, so that its rows summed to different constants, the sum would return $E^\top(p_1 \circ \mathrm{rowsum})$ instead and the gradient could leak *without* any confounding at all.

Confounding is exactly what breaks that sum. Table 2 maps both leakages across four configurations. In a *complete* design both are knife-edged at confounding 1. In an *incomplete* design β is essentially flat while β_g is **graded** in confounding, giving a floor that rises smoothly from 0 to 0.82 — eight non-degenerate cells with a nonzero floor.

4.3. The conjunction

Collecting the two switches:

	truth leaks ($\beta > 0$)	gradient leaks ($\beta_g > 0$)
$\lvert O_0\rvert < \lvert S\rvert$ (weak control)	**yes**	no
π_b depends on S (confounding)	no	**yes**

Corollary 3. *For a stage-1 bridge block, the floor of Proposition 1 is nonzero exactly when genuine confounding coincides with an inadequate negative control. Either condition alone leaves one factor at zero and the floor vanishes.*

The stage restriction is real, and it is good news. At $t = 1$ the conditioning set is (A_1, O_0), so a per-action design has only $|O_0|$ profiles and $|O_0| < |S|$ forces an incomplete span. From $t = 2$ on the conditioning set includes history: at $t = 2$ it is (A_2, O_1, A_1, O_0), giving $|O|\,|A|\,|O_0|$ cells per action, and the O_1-conditioned posteriors are generically rich enough to span all of $\mathbb{R}^{|S|}$. We checked this in the population: at every confounding level below 1, the $t = 2$ per-action span reaches $|S|$ and $\beta \approx 10^{-15}$ in $24/24$ cells, against a $t = 1$ span of 2 or 3 in the same configurations.

Corollary 4 (History is a partial substitute for the instrument)**.** *With $T > 1$ and confounding strictly below saturation, an inadequate negative control is* partially self-healing*: per-action completeness is restored at every stage after the first, and only the stage-1 blocks leak.*

The restoration is uniform rather than a $t = 2$ accident: we computed per-action spans stage by stage out to $t = 5$, and the span reaches $|S|$ at *every* stage after the first, at every confounding level below saturation. Only the first block leaks: across all configurations and stages $t \geq 2$ we measure $\beta_t \leq 2.7 \times 10^{-15}$ and a per-stage floor $\leq 4.3 \times 10^{-30}$.

The damage is diluted by the horizon. Because exactly one block leaks, the absolute floor does not compound: it is the same number whether the horizon is 1 or 6. The scale it must be compared against is not. Selection is decided by *differences* between candidate policies, and that spread grows with T. Table 3 divides one by the other and finds a decay of roughly $1/T$ — log–log slopes of -1.23 and -0.93, about a $10\times$ reduction from $T = 1$ to $T = 6$.

Corollary 5 (Horizon dilution)**.** *The floor is constant in T while the decision-relevant value spread grows linearly, so the fraction of the selection problem it distorts decays as* $\Theta(1/T)$.

Table 2. Population leakages and the resulting floor $\beta\beta_g$ against confounding strength. The floor is nonzero only where both switches are on.

config		0.0	0.3	0.6	0.9	0.99	1.0
$(2, 6, 4)$	β	0	0	0	0	0	.759
complete	β_g	0	0	0	0	0	.758
$(4, 6, 2)$	β	1.211	1.210	1.206	1.178	1.177	1.179
incomplete	β_g	0	.028	.093	.364	.678	.698
	floor	**0**	**.034**	**.112**	**.428**	**.798**	**.823**
$(4, 8, 3)$	**floor**	**0**	**.070**	**.125**	**.162**	**.171**	**.966**

Table 3. The floor is constant in T while the decision-relevant scale grows. Entries are $\beta\beta_g$ divided by the spread of true values over six candidate policies. Log–log slopes against T: -1.23 and -0.93.

config	conf.	T=1	2	3	4	5	6
$(4, 6, 2)$	0.6	0.425	0.161	0.099	0.072	0.056	0.046
$(4, 6, 2)$	0.9	**1.627**	0.615	0.379	0.274	0.215	0.177
$(4, 8, 3)$	0.6	0.390	0.205	0.141	0.108	0.088	0.074
$(4, 8, 3)$	0.9	0.505	0.266	0.183	0.140	0.114	0.096

Two consequences we state plainly because they cut in opposite directions. At $T = 1$ under strong confounding the floor *exceeds the entire value spread* (1.627): pessimism there cannot discriminate among candidates at all, because the irreducible penalty is larger than the whole range it would have to resolve. But by $T = 6$ the same quantity is 0.18. **Proposition 1 is horizon-free — the region does not contract at any T — while its decision-level consequence is a short-horizon phenomenon.** The regime in which proximal pessimism is genuinely unusable is short-horizon, strongly confounded, and weak-instrument, which is narrower than the width result alone would suggest.

5. Consequences for Rate Schedules

5.1. The power-law family

Specialize to $\lambda(N) = \lambda_0 N^{-\kappa}$ and $\xi(N) = c/(N\mu(N))$ with $\mu(N) = mN^{-\tau}$, and define the governing exponent

$$e \;:=\; \tau + \kappa - 1. \tag{5}$$

Corollary 6 (Dichotomy)**.** *While the bridge-class constraint is inactive:* (i) $W(N) = \Theta(N^{e/2})$*, with* $W(N) \geq \beta_g\sqrt{c/m}\,\lambda_0^{-1/2}N^{e/2}$*;* (ii) *coverage* requires $N^e \geq K$ *with* $K := \lambda_0\beta^2 m/c$*;* (iii) *if* $e < 0$ *the width contracts at* $N^{e/2}$ *but the necessary condition fails for all* $N > K^{1/e}$*, while if* $e \geq 0$ *the width is non-decreasing and never contracts.*

Two cautions, stated because earlier drafts of ours got them wrong. The constant K matters: for $e > 0$ the coverage condition fails for all $N < K^{1/e}$, so "$e \geq 0$ implies coverage at every N" needs $K \leq 1$. And (ii) is *necessary*, not sufficient — it says the null obstruction is absent, not that coverage holds. Signal directions demand width too, and on our toy design they dominate the required radius by $15\times$.

5.2. Where an existing schedule sits

The schedule of Hong et al. (2024) has width $\xi \sim M_R N_2^{-\alpha/(2\alpha+2)}$ and ridge $\lambda_2 = N_2^{-\alpha/(\alpha c_2+1)}$, which places it inside the family of Corollary 6 with

$$\tau = \frac{\alpha+2}{2\alpha+2}, \qquad \kappa = \frac{\alpha}{\alpha c_2 + 1},$$

and therefore

$$e \;=\; \frac{\alpha\,(2\alpha + 1 - \alpha c_2)}{(\alpha c_2+1)(2\alpha+2)}. \tag{6}$$

Their Assumption D.16 restricts c_2 to $(1, 2]$, so the factor $2\alpha + 1 - \alpha c_2$ is positive for every $\alpha > 0$. We verified Equation (6) against a direct evaluation of $\tau + \kappa - 1$ over the admissible grid: $e > 0$ in $\mathbf{30/30}$ cells, minimum 0.0217 at $\alpha = 10$, $c_2 = 2$.

Corollary 7. *The schedule of Hong et al. (2024) lies strictly on the non-contracting branch of Corollary 6 for every admissible smoothness. On a design where the floor is nonzero, the penalty it assigns therefore* grows *with* N.

We stress the conditional. Where that paper's assumptions hold, completeness forces $\beta = 0$ and Corollary 7 is decision-irrelevant: we measured $\beta_g = 0$ exactly for three policies at the exact bridges on a complete design, and four schedules including this one give 0.000 regret at every N tested. Corollary 7 is a statement about the boundary of applicability, and it is one the original analysis does not make.

5.3. What is not new here

Corollary 6 is not a new phenomenon. It is Tikhonov source-condition theory (Caponnetto & De Vito, 2007) specialized to a design with an exact null atom: in the classical setting with singular values $s_j \sim j^{-b}$ and gradient mass $g_j^2 \sim j^{-a}$, the same calculation gives $e' = \tau + \theta\kappa - 1$ with $\theta =$

$1-(a-1)/b<1$, and $\theta=1$ — our case — arises exactly when the spectrum has an atom at zero. Readers should treat Corollary 6 as the specialization it is.

Proposition 2 is also not new in substance. It is derivable from the anchor paper's own machinery: without confounding the behaviour policy is observable, so importance weights $\pi_e(a \mid o)/\pi_b(a)$ exist on observables for every observation-based policy, giving $C^*_\pi < \infty$, which by their Lemma C.1 forbids population gradient leakage. In proximal-inference terms it restates the folk fact that without unmeasured confounding a negative control is unnecessary. Our contribution here is the *conjunction* with the completeness switch and the observation that the two are independent, not either switch alone.

What we believe is new is Proposition 1, which is schedule-free and does not go through rates at all; Corollary 3 together with Corollary 4, which say which problems the floor applies to — and at which stage — in terms a practitioner can check; and Corollary 7, which locates a published schedule inside the dichotomy.

We also record a negative methodological finding about our own earlier work. The ratio of unprojected to projected width obeys, to machine precision over 17 cells of two independent sweeps,

$$\frac{W}{W_{\mathrm{proj}}} = \sqrt{1+\tan^2\theta\,\mathrm{cond}(H)}, \qquad \sin\theta = \beta_g, \tag{7}$$

an *identity*. Sweeping gradient alignment and observing that the width follows is therefore a re-measurement of arithmetic, not evidence. It also corrects a dichotomy we previously asserted: conditioning enters as $\sqrt{\mathrm{cond}(H)}$ on exactly equal footing with alignment, and the penalty is their product. Details in Appendix A.

6. A Decision-Level Consequence

A growing width matters only if it changes which policy is selected. We test that in a design where both switches are on, comparing pessimistic selection against a plug-in selector computed from the *same* fitted bridge inside the same loop, so that the only difference between the two columns is whether the pessimism layer is applied. All numbers are 20 seeds with 95% intervals.

Two readings of Table 4. First, the separation is complete: at $N = 128{,}000$ under schedule C, all 20 seeds put pessimism at 0.1804 and all 20 put the plug-in at the optimum, with zero interval on both, from a start where the two are statistically indistinguishable.

Second, and sharper, under the anchor paper's own schedule the curves **cross**. Pessimism is *better* at $N = 2{,}000$ and *worse* from $N = 32{,}000$ on. The layer helps when data is scarce, which is what it is for, and then actively hurts as data

Table 4. Selection regret, 20 seeds. Top: a schedule with $e = +0.50$. Bottom: the schedule of Hong et al. (2024), $e = +0.455$. Both switches on.

N	pessimistic	plug-in
Schedule C, $e=+0.50$		
2,000	0.0952 ± 0.0195	0.0894 ± 0.0194
8,000	0.1218 ± 0.0210	0.0226 ± 0.0138
32,000	0.1714 ± 0.0177	$\mathbf{0.0000 \pm 0.0000}$
128,000	$\mathbf{0.1804 \pm 0.0000}$	$\mathbf{0.0000 \pm 0.0000}$
Hong et al. (2024) schedule, $e=+0.455$		
2,000	$\mathbf{0.0694 \pm 0.0098}$	0.1641 ± 0.0000
8,000	0.0943 ± 0.0205	0.1641 ± 0.0000
32,000	0.1225 ± 0.0000	$\mathbf{0.0644 \pm 0.0000}$
128,000	0.1225 ± 0.0000	$\mathbf{0.0644 \pm 0.0000}$

accumulates. Regret slopes in $\log N$ confirm the pattern: under both schedules with $e > 0$ the pessimistic and plug-in slopes have *opposite signs* ($+0.0220$ against -0.0210 for schedule C, and $+0.0135$ against -0.0288 for the anchor schedule). More data converges the plug-in and diverges the pessimistic selection built on the same estimates.

7. Scope: What This Does and Does Not Establish

Does not. The binding hypothesis is $\beta\beta_g > 0$, and both halves live outside Hong et al. (2024)'s assumptions. Their completeness condition forces the min-norm bridge out of the population null, giving $\beta = 0$; and by their Lemma C.1, nonzero population gradient leakage is equivalent to an infinite concentrability coefficient C^*_π, which they exclude. **Nothing here is a defect in that paper.** Where its assumptions hold its construction behaves as advertised, and we measured exactly that.

Does. A robustness question the theory cannot ask of itself: what happens when the assumption it conditions on only approximately holds. The answer is that the failure is not graceful. It is not that the region contracts more slowly — it does not contract at all, and under an $e > 0$ schedule the penalty grows without bound while the estimate underneath it converges. Because the trigger is a conjunction of two checkable conditions, this is actionable rather than merely cautionary.

The exact-null assumption. Assumption 1 is exact for a tabular delta-kernel, where the design is a finite-rank count matrix. For a strictly positive-definite kernel the spectrum decays smoothly with no atom at zero, and Proposition 1 degrades to a statement about the effective null at the ridge scale rather than an exact one. We have not characterized that regime, and Section 5.3 indicates the exponent changes there.

8. Limitations

One environment family. All experiments are tabular POMDPs with $T \leq 3$ and $|S| \leq 4$, generated from a single construction. The population results are exact algebra and so do not depend on sampling, but they do depend on that family's structure.

The decision experiment sits at $T = 3$. By Corollary 5 the fraction of the selection problem the floor distorts decays as $1/T$, and $T = 3$ is a regime where that fraction is still 10–38$\%$. Table 4 is therefore a short-horizon result by construction. We have not run the 20-seed experiment at $T = 6$, where Table 3 predicts a materially weaker effect, and that is the sharpest open check on Section 6.

Dilution is measured, not derived. Corollary 5 rests on the value spread growing linearly in T, which holds in our environment family because per-stage rewards are bounded away from zero and candidate policies differ at every stage. A family with vanishing late-stage reward differences would dilute more slowly, and we have not characterized that.

Saturated confounding is a separate regime. All of Corollary 4 is stated for confounding strictly below 1. At exactly 1 the behaviour policy is a deterministic function of the latent state, history stops helping, and the $t = 2$ span collapses back to the $t = 1$ value. We regard that point as contrived, but it is the one place where the leak does propagate along the horizon.

Discrete regret. The decision experiment selects among six hand-constructed candidate policies, so regret takes few values. The crossing in Table 4 is stable across 20 seeds but has not been checked against a randomly generated candidate set.

Our own corrections. Several claims in earlier drafts of this line did not survive: a headline sample size computed on a grid that could not reach it, a validity result on a grid that could not falsify it, a decision sweep whose schedules all started inside an attractor, and the alignment result of Equation (7) that turned out to be an identity. Appendix C lists them.

9. Conclusion

Pessimism over a proximal bridge confidence region is only useful if the region contracts. We showed a floor that no width rule escapes, identified the exact conjunction of conditions under which that floor is nonzero, placed a published schedule strictly on the non-contracting branch, and exhibited a decision-level crossing where more data makes pessimistic selection worse while a plug-in built on the same estimates converges. The two triggering conditions are checkable before fitting, which turns a negative result into a pre-deployment test: if the negative control cannot resolve the latent state and the logging policy did, do not expect the region to shrink.

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A. The Width Identity, and What It Cost Us

Equation (7) deserves its own account because believing otherwise shaped an earlier version of this line of work.

Derivation. Under Assumption 1 the design has an exact null space on which H acts as λI, and a signal space on which its top eigenvalue is $\sigma + \lambda$. Splitting g into its in-span and null components, of squared norms $(1-\beta_g^2)\|g\|^2$ and $\beta_g^2\|g\|^2$,

$$\|g\|_{H^{-1}}^2 = \frac{(1-\beta_g^2)\|g\|^2}{\sigma+\lambda} + \frac{\beta_g^2\|g\|^2}{\lambda}, \qquad \|P_{\mathrm{span}} g\|_{H^{-1}}^2 = \frac{(1-\beta_g^2)\|g\|^2}{\sigma+\lambda},$$

whose ratio is $1 + \frac{\beta_g^2}{1-\beta_g^2} \cdot \frac{\sigma+\lambda}{\lambda}$. Writing $\sin\theta = \beta_g$ and $\mathrm{cond}(H) = (\sigma+\lambda)/\lambda$ gives Equation (7).

Verification. We checked the closed form against measurement over two independent knobs, 17 cells in total, and the worst relative error is 8.45×10^{-15} — machine precision at every point of both sweeps.

Why this matters. An earlier version of this work presented a sweep of gradient alignment against width ratio as the load-bearing empirical result, concluding that "ill-conditioning is necessary but not sufficient; the alignment between the value functional and the null space decides it". Both halves of that sentence are wrong as stated:

- The relation is an identity, so no sweep of it could ever have produced a different answer. It is arithmetic, not evidence.
- Conditioning enters as $\sqrt{\mathrm{cond}(H)}$ on *exactly equal footing* with $\tan\theta$. Neither factor "decides" the penalty; it is their product. The dichotomy was read off a sweep in which $\tan\theta$ moved $119\times$ while $\sqrt{\mathrm{cond}(H)}$ moved $1.26\times$ — a 95:1 imbalance built into the knob rather than a property of the problem.

The knob was also degenerate. That sweep varied alignment by interpolating every emission row toward their mean. That operation scales the row separation by exactly $(1-\alpha)$, and β_g tracks it with a log–log slope of 0.954 ($R^2 = 0.998$): alignment and channel informativeness were a single variable. At the far end of the sweep the emission channel carries almost no information about the latent state, so leakage falls for a reason unrelated to identification geometry. The claim that "the null space is held fixed" was also false — its *dimension* is fixed, but the span vector rotates $36.8°$ across the sweep.

A non-degenerate replacement. At confounding 1 with $|S| = |A| = 2$ the behaviour policy is a deterministic indicator, so the normalized stage-2 design weight sits on a single latent state whatever the prior is, making H *algebraically* independent of p_1. Tilting $p_1(t) = (1-t)p_1 + t\,\delta_{s_a}$ therefore moves alignment with the emission channel, the span and H all frozen:

t	row separation	$\max\lvert\Delta H\rvert$	span rotation	β_g	W/W_{proj}
0.000	0.81364	0	$0°$	0.7578	24.738
0.400	0.81364	5.6×10^{-17}	$0°$	0.4572	10.984
0.800	0.81364	5.6×10^{-17}	$0°$	0.1345	3.056
0.999	0.81364	5.6×10^{-17}	$0°$	0.00061	1.000

The same curve, 3.09 decades of β_g, at a channel that is never touched. So the qualitative conclusion survives on a clean knob; it simply was not being tested by the sweep that established it, and it is an identity in any case.

B. Experimental Setup

Environment family. Confounded POMDPs with independently specified $(|S|, |A|, |O|, |O_0|)$. Emission and negative-control matrices are drawn from a Dirichlet with a diagonal boost and asserted to have full row rank, so that identification never fails for an uninteresting reason. A confounding knob interpolates the behaviour policy between latent-state-independent and a deterministic function of the latent state. Horizon $T = 3$ throughout.

Population computations. Tables 1 and 2 involve no sampling. True bridges are minimum-norm solutions obtained by pseudo-inverse, and the design span for each action is the range of $\{E^\top v_{a,x}\}_x$ computed by SVD with a relative tolerance of 10^{-12}. Reported leakages are exact up to floating point; the maximum bridge residual across all cells is 4.4×10^{-16}.

Decision experiment. Six candidate policies, selection regret against the exact dynamic-programming optimum on the latent model. The plug-in selector is computed inside the same seed loop from the same fitted bridge, so the two columns of Table 4 differ only by the pessimism layer. Intervals are $1.96\hat{\sigma}/\sqrt{n}$ over 20 seeds.

A precondition guard. The decision driver aborts if $\beta_g < 10^{-6}$ on the configured grid. This exists because three earlier experiments in this line were run on grids that could not have exhibited the effect they were testing; see Appendix C.

C. Corrections to Our Own Results

We list the claims from earlier drafts of this work that did not survive, with what killed each. All were found by adversarial review of our own results rather than by external referees.

claim	disposition
A coverage crossing predicted at $N = 1.7 \times 10^9$	**withdrawn.** The grid reached 10^6. The prediction rested on treating a measured β as an environment constant; it in fact decays at $N^{-0.495}$ with a population value of 5.4×10^{-16}, so the quantity being extrapolated was a grid average of something converging to zero.
"Validity holds in $36/36$ cells"	**withdrawn as evidence.** The violation threshold sits four orders of magnitude below the smallest grid value. The grid could not have produced a violation.
"The width schedule changes the decision" (first sweep)	**refuted.** $\beta_g = 0$ exactly on that design, so there was no mechanism to detect, and three of four schedules started inside an attractor. Re-run on an incomplete design, where the floor is nonzero, it holds — Table 4.
"Gradient alignment is the causal driver, not conditioning"	**corrected.** The relation is the identity of Equation (7); conditioning enters as $\sqrt{\mathrm{cond}(H)}$ on equal footing. See Appendix A.
"The power-law dichotomy is novel"	**refuted.** It is Tikhonov source-condition theory specialized to an exact null atom (Section 5.3).
"A truth-dependent width rule would escape the floor"	**refuted.** Proposition 1 quantifies over all width rules including truth-dependent ones. The escapes are a prior-informed centre, a non-ellipsoidal region, or $\beta\beta_g = 0$.
"The plug-in also fails in the incomplete design"	**refuted before publication.** Measured at one value of κ only; because $\hat{b}$ depends on λ_2, that was not a control. Re-run at each schedule's own κ, the plug-in dominates.
"Schedule B is a flat $e = 0$ control"	**refuted.** A 3-seed artifact. At 20 seeds it runs $0.1230 \to 0.0644$ and plateaus.

Four of these share one failure mode: the measurement range did not straddle the transition being predicted. The precondition guard described in Appendix B is the structural fix for that mode. Two others — the degenerate alignment knob and the identity — are a different mode, in which the experiment could not have failed because its knob confounded two variables or its target was arithmetic. We have no structural fix for that one beyond deriving the closed form before sweeping.